

ROS 2

The framework used in real robots · Equigrow short notes

Core ideas

Concept	Meaning
Node	one program with one job (camera, planner...)
Topic	a named stream of messages (publish/subscribe)
Service	request-response call
Launch file	starts many nodes together
TF	keeps track of coordinate frames

Commands

```
ros2 run <package> <node>
ros2 topic list
ros2 topic echo /cmd_vel
ros2 launch <package> <file>.launch.py
colcon build && source install/setup.bash
```

A tiny publisher (Python)

```
import rclpy
from rclpy.node import Node
from std_msgs.msg import String
class Talker(Node):
    def __init__(self):
        super().__init__("talker")
        self.pub = self.create_publisher(String, "chatter", 10)
        self.create_timer(1.0, lambda: self.pub.publish(String(data="hi")))
rclpy.init(); rclpy.spin(Talker())
```